STANDARD OPERATING PROCEDURE (SOP) | UNIVERSAL INFRASTRUCTURE & DNS GUIDE

Vercel + WHM / cPanel DNS Setup

Evergreen Guide for Staging Subdomains & Production Domain Cutovers

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1. Overview & Understanding Vercel's Dynamic DNS Targets

When connecting a custom domain or staging subdomain (such as staging.<domain.com> or staging2.<domain.com>) to Vercel, Vercel generates project-specific DNS targets from its expanded global Anycast pool rather than relying solely on the legacy universal values.

Record Type	Legacy Vercel Value (Still Works)	Modern Project-Specific Value (Recommended)
CNAME (Subdomains)	cname.vercel-dns.com.	<unique-hash>.vercel-dns-xxx.com. (e.g. 18ab73975d5bfe50.vercel-dns-017.com.)
A Record (Apex @)	76.76.21.21	Expanded Anycast IP assigned in dashboard (e.g. 76.76.21.x or 216.198.79.x)

- Why the change? Vercel expanded its global IP and DNS infrastructure to improve routing resilience and isolate traffic across edge clusters.
- What should you use? Always copy and paste the exact Name and Value displayed inside your Vercel project's Settings -> Domains screen. Both the modern project-specific record and the legacy record work, but Vercel's dashboard validates fastest when matching the exact recommended record shown.

2. Golden Rule: Zone Editor Only (Never Create a cPanel Subdomain)

When your DNS is hosted on a WHM/cPanel server (which may also host your current website, Headless WordPress, or email) and your frontend application is hosted externally on Vercel:

- DO NOT go to cPanel -> Domains -> Create A New Domain to create your staging subdomain.
  - Why? Creating a domain/subdomain inside cPanel's "Domains" interface tells the cPanel server to host the files locally and automatically creates local A records pointing to the cPanel server's own IP address.
- DO go directly to cPanel -> Zone Editor (or WHM -> DNS Zone Manager).
  - You only need a DNS record (CNAME or A) pointing traffic directly to Vercel's servers.

3. Part 1: Configuring the Domain or Subdomain in Vercel

Before touching WHM or cPanel, register the target hostname inside your Vercel project so Vercel knows which project should serve incoming requests and gives you your exact DNS target.

1. Log in to Vercel and open your project.
2. Click Settings in the top navigation bar, then select Domains from the left sidebar.
3. In the domain input box, enter your target domain:
   - For a Staging Site: Enter <subdomain>.<domain.com> (e.g., staging2.goodrep.media).
   - For a Production Launch: Enter <domain.com> (and optionally let Vercel add www.<domain.com>).
4. Click Add.
5. Vercel will immediately display an "Invalid Configuration" red badge. This is normal because the DNS record has not been added to WHM/cPanel yet.
6. Ensure the DNS Records tab is selected (not Vercel DNS, since WHM/cPanel manages your nameservers).
7. Copy the exact Name and Value provided by Vercel:

```
+-------------------------------------------------------------------------------+
|  VERCEL DASHBOARD -> SETTINGS -> DOMAINS                                      |
|                                                                               |
|  Type        Name               Value                                         |
|  ----        ----               -----                                         |
|  CNAME       <subdomain>        <unique-hash>.vercel-dns-017.com.             |
|  (Example)   staging2           18ab73975d5bfe50.vercel-dns-017.com.          |
+-------------------------------------------------------------------------------+
```

4. Part 2: Adding the DNS Record in cPanel (Zone Editor)

Use this workflow when logged into the individual cPanel account for <domain.com>.

Step 4.1: Open Zone Editor & Check for Conflicts

1. Log in to cPanel.
2. Scroll to the Domains section and click Zone Editor.
3. Locate <domain.com> in the table and click Manage (on the far right).
4. In the Filter / Search bar above the records table, type your subdomain prefix (e.g., staging2).
5. Inspect existing records:
   - DNS rules strictly forbid a CNAME record from coexisting with any A, AAAA, or TXT record of the same name.
   - If you see any existing A or CNAME record for <subdomain>.<domain.com>., click Delete (and also delete any auto-generated mail.<subdomain>, cpanel.<subdomain>, or webmail.<subdomain> records if someone previously added it as a cPanel domain).

Step 4.2: Create the CNAME Record

1. Click the + Add Record button (or click the dropdown arrow next to + Add Record and select Add "CNAME" Record).
2. Complete the fields exactly as follows:

cPanel Zone Editor Field	What to Enter	Example (staging2.goodrep.media)
Name (Zone Name)	Enter <subdomain> (cPanel will automatically append .<domain.com>. when you click or tab out of the box)	staging2 (becomes staging2.goodrep.media.)
TTL	300 (or leave default 14400; 300 = 5 mins for fast propagation)	300
Type	CNAME	CNAME
Record (Target)	Paste the Value copied from Vercel (with or without trailing dot; cPanel normalizes it)	18ab73975d5bfe50.vercel-dns-017.com.

3. Click Save Record.

5. Part 3: Adding the DNS Record in WHM (DNS Zone Manager)

Use this workflow if you are logged into WHM (WebHost Manager) as a root/reseller administrator rather than an individual cPanel user.

1. Log in to WHM.
2. In the left navigation search bar, type DNS and click DNS Functions -> DNS Zone Manager.
3. Find <domain.com> in the domain list and click Manage.
4. Use the search input to check if <subdomain>.<domain.com>. already exists. If conflicting A or CNAME records exist for that exact subdomain, click Delete.
5. Click + Add Record at the top right and select CNAME:
   - Valid Zone Name: Enter <subdomain> (WHM automatically appends .<domain.com>.).
   - TTL: 300
   - Type: CNAME
   - Record (Hostname): Paste the exact Vercel value (e.g., 18ab73975d5bfe50.vercel-dns-017.com.).
6. Click Save Record.

6. Part 4: Universal Reference Matrix (Staging vs. Production Cutover)

When managing client sites in WHM/cPanel, you will typically perform one of two operations: setting up an isolated Staging Subdomain (zero impact on the live site), or executing a Full Production Cutover (where Vercel takes over <domain.com> and www.<domain.com>, while WordPress and Email remain on the WHM/cPanel server).

Scenario A: Staging / Preview Subdomain Only (Zero Live Impact)

Leaves the current live website (@ and www) and email completely untouched on WHM/cPanel:

Type	Name (in Zone Editor)	Target / Record Value	TTL	Purpose
CNAME	staging2.<domain.com>.	<hash>.vercel-dns-xxx.com.	300	Routes https://staging2.<domain.com> to Vercel

Scenario B: Full Production Cutover (Apex + www on Vercel, WordPress + Email on cPanel)

When moving the live domain to Vercel while keeping Headless WordPress (wp.<domain.com>) and cPanel Email on your WHM server:

Type	Name (in Zone Editor)	Target / Record Value	TTL	Purpose
A	wp.<domain.com>.	<cpanel-server-ip>	300	Keeps Headless WordPress CMS hosted on cPanel server
A	mail.<domain.com>.	<cpanel-server-ip>	300	Keeps cPanel mail server reachable (if using cPanel email)

Type	Name (in Zone Editor)	Target / Record Value	TTL	Purpose
MX	<domain.com>.	Priority: 0 -> mail.<domain.com>.	300	Ensures MX points to mail. (NOT @) before changing @ IP
A	<domain.com>.	<vercel-assigned-a-record-ip>	300	Routes root domain (https://<domain.com>) to Vercel
CNAME	www.<domain.com>.	<hash>.vercel-dns-xxx.com.	300	Routes https://www.<domain.com> to Vercel

- CRITICAL EMAIL CHECK FOR PRODUCTION CUTOVERS: If a client uses cPanel Webmail/Email (instead of Google Workspace or Microsoft 365), their default MX record often points directly to <domain.com>.. Before pointing the root <domain.com>. A record to Vercel's IP, verify that the MX record points to mail.<domain.com>. and that mail.<domain.com>. has an A record pointing to the cPanel server IP.

7. Part 5: Verification, SSL Issuance & Troubleshooting

Step 7.1: Verify DNS Propagation & SSL in Vercel

1. Wait 1 to 5 minutes after saving the record in WHM/cPanel.
2. Return to your Vercel Dashboard -> Settings -> Domains.
3. Vercel automatically polls the DNS record every few seconds (or you can click Refresh).
4. Once detected:
   - The red Invalid Configuration badge changes to Valid Configuration (blue checkmark).
   - Vercel automatically provisions and installs a free Let's Encrypt SSL/TLS certificate in the background (usually within 30-60 seconds).
5. Open https://<subdomain>.<domain.com> in your browser to confirm your site is live over HTTPS.

Step 7.2: Optional Terminal Check

You can test DNS resolution immediately from your terminal:

```
# Verify the CNAME record resolves to Vercel's target
dig CNAME staging2.goodrep.media +short

# Expected output:
# 18ab73975d5bfe50.vercel-dns-017.com.
```

Step 7.3: Common Troubleshooting Fixes

Issue / Error Message	Root Cause	Quick Fix
cPanel Error: "A CNAME record with that name already exists or conflicts with another record"	An A, AAAA, or TXT record already exists for <subdomain>.<domain.com>. in the Zone Editor.	Filter the Zone Editor by <subdomain>, delete the existing A / TXT records for that subdomain, then save the CNAME.
Browser shows cPanel "Default Web Site Page" (cgi-sys/defaultwebpage.cgi)	The subdomain was created under cPanel -> Domains OR local browser/ISP DNS cache is still holding the old IP.	1. Remove the subdomain from cPanel -> Domains if present. / 2. Re-add the CNAME in Zone Editor. / 3. Test in an Incognito window or flush DNS cache.
Vercel stuck on "Generating SSL Certificate" for >15 minutes	The domain has restrictive CAA records in cPanel blocking Let's Encrypt.	In cPanel Zone Editor, filter by CAA. Either remove restrictive CAA records or add a CAA record permitting letsencrypt.org.